



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
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Roll Number	2520090205
Branch	CSIT
Course Outcome	4

Case Study Topic: SecureVote – Digital Voting & Election Analytics System using Bellman-Ford Algorithm

Work Done Summary:

Implemented the **Bellman-Ford Algorithm** in Java for election monitoring and voter network analysis. Modeled voters, polling stations, and election activities as graph nodes connected through weighted edges. Bellman-Ford traversal was used to calculate the shortest communication paths between election centers and polling stations. Implemented voter registration, authentication verification, ballot casting, vote counting, election analytics, and fraud detection using graph-based shortest path techniques. Demonstrated shortest path computation for auditing election data flow and monitoring connectivity among election nodes. Compared Bellman-Ford with Dijkstra's Algorithm – Bellman-Ford is more suitable when negative edge weights may exist and can also detect negative weight cycles indicating suspicious network activity.

Implementation Details :

1. Created Graph class using Edge representation.
2. Defined Edge class with source, destination, and weight fields.
3. BellmanFord(source) computes shortest distances from the source node to all other nodes.
4. Relaxed all edges repeatedly for **V-1** iterations.
5. Performed an additional iteration to detect negative weight cycles.
6. Election nodes represent polling stations, servers, and voter centers.
7. Fraud detection performed by identifying abnormal negative cycles in the election network.
8. Main method created election graph and demonstrated shortest path analysis using Bellman-Ford Algorithm.

Result : Screenshots/Output

```
PS D:\JavaPrograms> javac SecureVoteBellmanFord.java
PS D:\JavaPrograms> java SecureVoteBellmanFord
SecureVote - Digital Voting & Election Analytics System
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Election Network Connections:
0 -> 1 (Weight 4)
0 -> 2 (Weight 2)
1 -> 2 (Weight 3)
1 -> 3 (Weight 2)
1 -> 4 (Weight 3)
2 -> 1 (Weight 1)
2 -> 3 (Weight 4)
3 -> 4 (Weight 2)

Shortest Distance from Main Election Server:
Node 0 -> 0
Node 1 -> 3
Node 2 -> 2
Node 3 -> 5
Node 4 -> 6

Fraud Detection:
No Fraudulent Activity Detected.
PS D:\JavaPrograms>
```

Time Complexity:

- Bellman-Ford Traversal: $O(V \times E)$
- Graph Construction: $O(E)$
- Bellman-Ford efficiently computes shortest paths and detects negative cycles in election monitoring networks.
- **GitHub Link :** https://github.com/anumolajhansi08-sketch/DSA-2_CO4

Student Signature	Faculty Signature